JEE Mathematics: Quick Review with Grading and Practice Map - Parabola (Batch XII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (84 questions; a question that appears in two of these is counted once): High = 8 or more questions, Medium = 1 to 7 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Conic Section : Definition

Question 01

What is the definition of conic section?

Model Answer

Locus of a point P moving in a plane so that $\frac{PS}{PM} = e$ (constant), where S is a fixed point and PM is the perpendicular distance of P from a fixed line.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 02

Define focus, directrix and eccentricity for a conic.

Model Answer

Focus = the fixed point S; directrix = the fixed line; eccentricity $e = \frac{PS}{PM}$ = the constant ratio.

$e = 1$: parabola; $0 < e < 1$: ellipse; $e > 1$: hyperbola.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 03

Define axis, vertex and foot of directrix for a conic.

Model Answer

Axis: line through the focus perpendicular to the directrix. Vertex: point where the conic meets its axis. Foot of directrix Z: point where the axis meets the directrix.

For a parabola the vertex is the mid-point of S and Z.

Grade - Low

Practice map, main concept: none

Also used in: none

General Equation of Conic

Question 04
What is the general form of equation of conic?

Model Answer

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$$

It comes from $PS = e\,PM$ with $S(\alpha, \beta)$ and directrix $lx+my+n = 0$: $(l^2+m^2)\left[(x-\alpha)^2 + (y-\beta)^2\right] = e^2(lx + my + n)^2$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 05
How to determine nature of conic if equation of conic is given?

Model Answer

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

$\Delta = 0$: pair of lines (real if $h^2 > ab$, parallel/coincident if $h^2 = ab$). $\Delta \neq 0$: parabola if $h^2 = ab$; ellipse if $h^2 < ab$ (circle if $a = b,\ h = 0$); hyperbola if $h^2 > ab$ (rectangular if $a + b = 0$).
Ex: $4x^2 - 4xy + y^2 - 12x + 6y + 9 = 0$: $h^2 = ab$ but $\Delta = 0$, it is $(2x - y - 3)^2 = 0$ (coincident lines, not a parabola).

Grade - **Low**
Practice map, main concept: Sheet: J-II 95
Also used in: none

Question 06
How to determine nature of conic if focus, eccentricity and directrix is given?

Model Answer
Focus on the directrix: pair of lines (real and distinct if $e > 1$, coincident if $e = 1$, imaginary if $e < 1$).
Focus not on the directrix: $e = 1$ parabola, $e < 1$ ellipse, $e > 1$ hyperbola.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 07
How to observe focus, e and directrix if equation is given in $PS = e\,PM$ form?

Model Answer

$$\sqrt{(x-\alpha)^2 + (y-\beta)^2} = e \cdot \frac{|lx + my + n|}{\sqrt{l^2 + m^2}}$$

Make the right side a true perpendicular distance (divide by $\sqrt{l^2 + m^2}$); then focus (α, β), directrix $lx + my + n = 0$, and e is the factor left.
Ex: $(x-3)^2+(y+2)^2 = \frac{1}{5}(3x+4y-3)^2$ is $PS = \sqrt{5}\cdot\frac{|3x+4y-3|}{5}$: focus $(3, -2)$, directrix $3x+4y-3 = 0$, $e = \sqrt{5}$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 1(5)

Parabola : Introduction

Question 08

What is the condition for general equation of 2nd degree curve to represent a parabola?

Model Answer

$\Delta \neq 0$ and $h^2 = ab$ (then $e = 1$)

Ex: $x^2 + 2xy + y^2 - 2x - 1 = 0$: $h^2 = ab = 1$, $\Delta = -1 \neq 0$, so it is a parabola.

Grade - Low

Practice map, main concept: DPP: 1(9)

Also used in: Sheet: J-II 95

Question 09

What is the definition of parabola?

Model Answer

Locus of a point whose distance from a fixed point (focus) equals its perpendicular distance from a fixed line (directrix): $PS = PM$, i.e. $e = 1$.

Focus $(a, 0)$, directrix $x + a = 0$ give $y^2 = 4ax$.

Grade - Low

Practice map, main concept: Sheet: C-II 63

Also used in: none

Question 10

Define focal distance/focal radius, focal chord, double ordinate, latus rectum for a parabola

Model Answer

Focal distance of P: PS (= distance of P from the directrix). Focal chord: chord through S. Double ordinate: chord perpendicular to the axis. Latus rectum: the double ordinate through S.

Ex: given the two ends of a latus rectum, two parabolas are possible (opening on either side).

Grade - Low

Practice map, main concept: DPP: 1(8, 10); 9(17)

Also used in: none

Question 11

Everything means focus, directrix, vertex, axis, ends of latus rectum, length of latus rectum (LR).

Model Answer

For every parabola list: vertex, focus, directrix, axis, tangent at vertex, foot of directrix, equation of LR, ends of LR and length of LR.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 12

Find everything for standard parabola $y^2 = 4ax$.

Model Answer

Vertex $(0, 0)$; focus $(a, 0)$; directrix $x + a = 0$; axis $y = 0$; tangent at vertex $x = 0$; foot of directrix $(-a, 0)$; LR: $x = a$, ends $(a, \pm 2a)$, length $4a$.

Ex: $y^2 = 12x$: $a = 3$, focus $(3, 0)$, directrix $x = -3$, LR $= 12$.

Grade - Medium

Practice map, main concept: Sheet: J-M 14; J-A 1, 19

Also used in: Sheet: J-II 70; J-A 20; C-I 13; C-II 31

Question 13

What is the relation between length of LR and distance between focus and corresponding directrix for a parabola?

Model Answer

LR $= 4a = 2\times$ (distance of focus from directrix)

Also vertex to focus $= a =$ vertex to directrix. Ex: vertex $(2, -1)$, directrix $4x - 3y = 21$: distance $= \frac{|8+3-21|}{5} = 2$, so LR $= 8$.

Grade - Medium

Practice map, main concept: Sheet: J-M 29, 50, 62 | DPP: 1(7)

Also used in: Sheet: J-II 93

Question 14

Find everything for parabola if equation is given in $PS = e\,PM$ form.

Model Answer

Read focus S and directrix; axis: line through S perpendicular to the directrix; $Z =$ axis $\cap$ directrix; vertex = mid-point of SZ; LR $= 2\times$ distance of S from directrix; tangent at vertex $\parallel$ directrix.

Ex: $(x-1)^2 + (y-1)^2 = \left(\frac{x+y-6}{\sqrt{2}}\right)^2$: $S(1,1)$, $Z(3,3)$, vertex $(2,2)$, axis $x = y$, LR $= 4\sqrt{2}$, tangent at vertex $x + y = 4$.

Grade - Medium

Practice map, main concept: Sheet: J-II 61, 69, 84; C-II 55 | DPP: 1(5-6) | Question Bank: 73, 79

Also used in: Sheet: J-II 87; J-M 28; C-I 10 | DPP: 1(8) | Question Bank: 72

Question 15

What is the locus of center of a variable circle which touches a fixed circle and a fixed line?

Model Answer

A parabola: focus = centre C of the fixed circle (radius r), directrix = the fixed line shifted by r away from C (external touching; towards C for internal).

Ex: touching $x^2 + y^2 = 1$ externally and $y = -2$: centre lies on $x^2 = 6y + 9$ (focus O, directrix $y = -3$).

Grade - Low

Practice map, main concept: Sheet: J-II 68 | DPP: 1(1)

Also used in: Sheet: J-II 84

Question 16

How to find focal distance of a point on a parabola/conic?

Model Answer

$PS = e \cdot PM$ (distance from the directrix). For $y^2 = 4ax$: $PS = x_1 + a = a(1 + t^2)$.

Ex: $y^2 = 8x$, $P(2,4)$: $PS = 2 + 2 = 4$.

Grade - Medium

Practice map, main concept: Sheet: J-II 85; C-II 54

Also used in: Sheet: J-II 83; J-M 25; J-A 37 | Question Bank: 75

Other standard parabolas

Question 17
What are three other standard parabolas?

Model Answer

$$y^2 = -4ax, \quad x^2 = 4ay, \quad x^2 = -4ay \quad (a > 0)$$

All have vertex O and LR $= 4a$; they open to the left, upwards and downwards respectively.

Grade - Medium
Practice map, main concept: Sheet: J-I 2; J-II 73
Also used in: Sheet: J-M 23 | DPP: 3(10)

Question 18
Plot all 4 standard parabolas and mark everything for them.

Model Answer
$y^2 = 4ax$: $S(a, 0)$, $x = -a$. $y^2 = -4ax$: $S(-a, 0)$, $x = a$. $x^2 = 4ay$: $S(0, a)$, $y = -a$. $x^2 = -4ay$: $S(0, -a)$, $y = a$.
The curve always bends round the focus; directrix is on the other side of the vertex; LR is the line through S parallel to the directrix.

Grade - Low
Practice map, main concept: none
Also used in: none

Shifted Parabola

Question 19
How is the equation of a shifted parabola represented?

Model Answer

$$(y - k)^2 = 4a(x - h), \qquad (x - h)^2 = 4a(y - k)$$

Vertex (h, k); axis parallel to the x-axis (first) or y-axis (second); $a < 0$ reverses the opening.

Grade - Low
Practice map, main concept: Sheet: C-II 15
Also used in: Sheet: C-II 19

Question 20
How to observe whether equation of a 2nd degree curve represents shifted parabola?

Model Answer
No xy term and only one square term: $y^2 + Dx + Ey + F = 0$ $(D \neq 0)$ or $x^2 + Dx + Ey + F = 0$ $(E \neq 0)$. Complete the square.
Ex: $y = x^2 - 4x$ is $(x - 2)^2 = y + 4$: vertex $(2, -4)$, $4a = 1$.

Grade - Medium
Practice map, main concept: Sheet: J-M 25; C-I 1; C-II 65 | DPP: 1(2)
Also used in: Sheet: C-I 14

Question 21
Find everything for shifted parabola.

Model Answer
Put $X = x - h$, $Y = y - k$, read everything for $Y^2 = 4aX$ and shift back. For $(y-k)^2 = 4a(x-h)$: focus $(h+a, k)$, directrix $x = h - a$, axis $y = k$, LR ends $(h+a, k \pm 2a)$, LR $= 4|a|$.

Ex: $y^2 - 4y + 16x - 44 = 0 \Rightarrow (y-2)^2 = -16(x-3)$: vertex $(3,2)$, $a = -4$, focus $(-1,2)$, directrix $x = 7$, LR $= 16$.

Grade - High

Practice map, main concept: Sheet: J-I 1, 33; J-II 50; J-M 33; J-A 11 | DPP: 1(3, 11); 2(5); 5(5) | Question Bank: 82
Also used in: Sheet: J-II 27; J-M 2, 13, 15, 51-52; J-A 34-35; C-I 6 | DPP: 4(11)

Question 22
How to find equation of shifted parabola if some details about it are given?

Model Answer
Axis parallel to a coordinate axis: take $(y-k)^2 = 4a(x-h)$ or $(x-h)^2 = 4a(y-k)$ and use the data (vertex, focus, a, a point). Oblique axis: use $PS = PM$ (or the alternate definition).

Ex: vertex $(3,2)$, focus $(3,6)$: $a = 4$, axis $\parallel$ y-axis, so $(x-3)^2 = 16(y-2)$.

Grade - Medium

Practice map, main concept: Sheet: J-II 87, 93; J-M 28, 31; C-I 10; C-II 29 | Question Bank: 72
Also used in: Sheet: J-II 50; J-M 29, 50 | DPP: 1(10); 9(17)

Alternate definition of Parabola

Question 23
Derive alternate definition of parabola.

Model Answer
In $y^2 = 4ax$, $|y|$ = distance from the axis and x = distance from the tangent at vertex: (distance from axis)^2 = LR $\times$ (distance from tangent at vertex).

$$\left(\frac{lx+my+n}{\sqrt{l^2+m^2}}\right)^2 = 4a \cdot \frac{mx-ly+n'}{\sqrt{l^2+m^2}}$$

Grade - Low

Practice map, main concept: none
Also used in: none

Question 24
Find everything for parabola if equation is given in alternate definition form.

Model Answer
With $L_1 \perp L_2$ normalised: axis $L_1 = 0$, tangent at vertex $L_2 = 0$, vertex $= L_1 \cap L_2$, LR $= 4a$; focus on the axis at distance a from the vertex on the side where $L_2 > 0$.

Ex: tangent at vertex $3x - 4y = 5$, focus $(1,2)$: $a = 2$, axis $4x + 3y = 10$, parabola $\left(\frac{4x+3y-10}{5}\right)^2 = 8 \cdot \frac{4y-3x+5}{5}$.

Grade - Low

Practice map, main concept: none
Also used in: none

Parametric Coordinates

Question 25

Write parametric coordinates of a point on standard parabolas.

Model Answer

$y^2 = 4ax$: $(at^2, 2at)$; $y^2 = -4ax$: $(-at^2, 2at)$; $x^2 = 4ay$: $(2at, at^2)$; $x^2 = -4ay$: $(2at, -at^2)$.

Locus questions: take the moving point as $(at^2, 2at)$, find (h, k) in terms of t, eliminate t. Ex: mid-point of $(1, 0)$ and a point of $y^2 = 8x$ lies on $y^2 = 4x - 2$.

Grade - Medium

Practice map, main concept: Sheet: J-I 3-5; J-II 49, 83; C-I 6, 13; C-II 9 | DPP: 4(9) | Question Bank: 88

Also used in: Sheet: J-I 6, 17; J-II 18, 21-22, 79; J-A 6, 13; C-I 2; C-II 12, 16, 54 | DPP: 1(12); 2(1, 3)

Question 26

What is the slope of radius vector of a point P in terms of t?

Model Answer

$$m_{OP} = \frac{2at}{at^2} = \frac{2}{t}$$

Grade - Low

Practice map, main concept: Sheet: J-I 6

Also used in: Sheet: J-I 19

Question 27

How to assume parameters for ends of double ordinate?

Model Answer

t and $-t$: $(at^2, \pm 2at)$.

Ex: points of trisection of double ordinates of $y^2 = lx$ lie on $9y^2 = lx$ (LR $\frac{l}{9}$).

Grade - Medium

Practice map, main concept: DPP: 1(12)

Also used in: Sheet: J-II 8, 89 | Question Bank: 78

Question 28

What is the parameter for ends of LR?

Model Answer

$t = \pm 1$: $(a, \pm 2a)$. Tangents at the ends meet at $(-a, 0)$, normals at $(3a, 0)$; together they form a square.

Grade - Medium

Practice map, main concept: Sheet: J-II 8; J-A 31

Also used in: Sheet: J-I 20; J-A 12 | DPP: 1(4); 3(15); 4(11-12); 6(13)

Question 29

What is the slope of line joining two points t_1 and t_2?

Model Answer

$$m = \frac{2}{t_1 + t_2}$$

Grade - Medium

Practice map, main concept: Sheet: J-II 79

Also used in: Sheet: J-A 9 | DPP: 3(12)

Question 30
What is the relation between parameters of ends of a chord subtending right angle at vertex?

Model Answer

$$t_1 t_2 = -4$$

From $\frac{2}{t_1} \cdot \frac{2}{t_2} = -1$.

Grade - **Medium**
Practice map, main concept: Sheet: J-II 86; J-A 13
Also used in: Sheet: J-II 26, 44 | DPP: 1(16)

Question 31
What is length of focal chord in terms of parameter?

Model Answer
Length $= a\left(t + \dfrac{1}{t}\right)^2$
Segments: $SP = a(1 + t^2)$, $SQ = a\left(1 + \frac{1}{t^2}\right)$. Ex: $y^2 = 4x$, $t = 2$: length $= \frac{25}{4}$.

Grade - **Medium**
Practice map, main concept: Sheet: J-A 4
Also used in: Sheet: J-M 38 | Question Bank: 83

Question 32
What is the relation between parameters of ends of a focal chord?

Model Answer

$$t_1 t_2 = -1$$

Ex: one end of a focal chord of $y^2 = 32x$ is $(2, -8)$ $(t = -\frac{1}{2})$: other end $t = 2$, i.e. $(32, 32)$.

Grade - **High**
Practice map, main concept: Sheet: J-M 3, 38, 63; J-A 9; C-I 8 | DPP: 1(14); 2(3) | Question Bank: 71, 75, 78
Also used in: Sheet: J-M 30; J-A 4-5, 10 | DPP: 3(8) | Question Bank: 70

Question 33
What is length of focal chord whose inclination is α?

Model Answer
Length $= 4a\,\mathrm{cosec}^2\alpha$
Least $=$ LR $4a$ $(\alpha = 90^\circ)$; distance of this chord from the vertex $= a\sin\alpha$. Ex: $y^2 = 12x$, slope $\sqrt{3}$: length $= 12 \cdot \frac{4}{3} = 16$.

Grade - **Medium**
Practice map, main concept: Sheet: C-I 15 | DPP: 1(13, 15, 17) | Question Bank: 84-85
Also used in: Sheet: C-II 17

Question 34
Write S, S_1 and T for parabola $y^2 = 4ax$.

Model Answer

$$S \equiv y^2 - 4ax, \quad S_1 \equiv y_1^2 - 4ax_1, \quad T \equiv yy_1 - 2a(x + x_1)$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Position of a point wrt Parabola

Question 35

Which region is called inside/outside wrt a parabola?

Model Answer

Inside: the region containing the focus (bounded by the curve on the concave side). Outside: the region containing the directrix; tangents can be drawn only from outside points.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 36

How to find relative position of a point wrt a parabola?

Model Answer

$S_1 = y_1^2 - 4ax_1$: $S_1 < 0$ inside, $S_1 = 0$ on, $S_1 > 0$ outside.

Ex: $(2, 5)$ and $y^2 = 8x$: $S_1 = 25 - 16 = 9 > 0$, outside.

Grade - Low

Practice map, main concept: DPP: 2(14-15)

Also used in: none

Chord joining two points

Question 37

Write expression of equation of chord joining two points t_1 and t_2.

Model Answer

$$2x - (t_1 + t_2)y + 2at_1t_2 = 0$$

Grade - Medium

Practice map, main concept: Question Bank: 89

Also used in: Sheet: J-I 3; C-II 13

Question 38

What is the relation between parameters of end points of a chord if chord passes through $(c, 0)$?

Model Answer

$$t_1t_2 = -\frac{c}{a}$$

$c = a$ gives the focal chord $t_1t_2 = -1$; $c = 4a$ gives $t_1t_2 = -4$.

Grade - Low

Practice map, main concept: Sheet: C-II 13

Also used in: Sheet: J-II 14; C-II 62

Question 39
What is the relation between parameters of end points of a chord if chord subtends right angle at vertex?

Model Answer

$$t_1 t_2 = -4$$

Then the chord $2x - (t_1 + t_2)y - 8a = 0$ always cuts the axis at $(4a, 0)$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 40
The chord subtending right angle at vertex always passes through which fixed point?

Model Answer

$$(4a, 0)$$

Ex: for $y^2 = 12x$ the fixed point is $(12, 0)$; feet of perpendiculars from O on such chords lie on $x^2 + y^2 - 4ax = 0$.

Grade - Low
Practice map, main concept: Sheet: J-II 44; C-II 62 | DPP: 1(16)
Also used in: Sheet: J-II 86; C-II 18

Question 41
What is the formula for length of chord intercepted by $y = mx + c$ on $y^2 = 4ax$?

Model Answer
Length $= \dfrac{4}{m^2}\sqrt{a(1+m^2)(a-mc)}$

Real chord needs $a > mc$. Ex: $y^2 = 4x$, $y = x - 1$ (focal, 45°): $4\sqrt{1 \cdot 2 \cdot 2} = 8 = 4a\,\mathrm{cosec}^2 45°$.

Grade - Medium
Practice map, main concept: Sheet: C-II 57 | DPP: 2(1-2)
Also used in: Sheet: J-A 44 | DPP: 4(6)

Line and Parabola

Question 42
How to find angle subtended by a line at the vertex of parabola? and how to check whether it is acute or obtuse?

Model Answer
Homogenise: write the line as $1 = \frac{lx+my}{-n}$ and put it in $y^2 = 4ax \cdot 1$ to get $Ax^2 + 2Hxy + By^2 = 0$; $\tan\theta = \dfrac{2\sqrt{H^2 - AB}}{|A+B|}$; $A + B = 0$: right angle, $A + B > 0$: acute, $A + B < 0$: obtuse.

Ex: $3x - 2y + 12 = 0$ and $4y = 3x^2$: $9x^2 + 3xy - 2y^2 = 0$, $\tan\theta = \frac{9}{7}$ (acute).

Grade - Medium
Practice map, main concept: Sheet: J-II 14; J-A 5 | Question Bank: 76
Also used in: Sheet: J-II 88

Question 43
What is condition of tangency for parabola $y^2 = 4ax$? And how is it used?

Model Answer

$$c = \frac{a}{m}$$

So $y = mx + \dfrac{a}{m}$ touches $y^2 = 4ax$ for every $m \neq 0$.
Use it for tangents with a given slope, from a point, common tangents and locus questions. Ex: $y = 3x + c$ touches $y^2 = 12x$ when $c = 1$.

Grade - High
Practice map, main concept: Sheet: J-M 21, 39; J-A 14; C-II 8
Also used in: Sheet: J-I 35; J-II 20, 42, 88; J-M 10, 24, 32; J-A 8, 25-26, 41; C-I 7, 9; C-II 10 | DPP: 2(9, 16); 4(3); 6(3); 7(8); 10(7)

Director Circle

Question 44
What is director circle for a parabola?

Model Answer
Locus of the point of intersection of perpendicular tangents; for a parabola it is the directrix $x + a = 0$.
Ex: for $y^2 = 16(x - 3)$ it is $x = -1$.

Grade - Medium
Practice map, main concept: Sheet: J-M 13 | DPP: 2(12)
Also used in: none

Question 45
Derive equation of director circle of parabola $y^2 = 4ax$.

Model Answer
$y = mx + \frac{a}{m}$ through (h, k): $hm^2 - km + a = 0$; $m_1 m_2 = \frac{a}{h} = -1 \Rightarrow h = -a$, i.e. $x + a = 0$.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 46
How to find locus of a variable external point P for which condition on slope of tangents drawn from P to parabola/2nd degree curve is given?

Model Answer
Put $P(h, k)$ in the slope form of the tangent: $hm^2 - km + a = 0$, so $m_1 + m_2 = \frac{k}{h}$, $m_1 m_2 = \frac{a}{h}$; write the condition with these and replace (h, k) by (x, y) (same idea for any curve with its tangency condition).
Ex: $y^2 = 4x$, $\theta_1 + \theta_2 = \frac{\pi}{4}$: $\frac{k/h}{1-1/h} = 1 \Rightarrow x - y - 1 = 0$.

Grade - Low
Practice map, main concept: DPP: 2(9, 16)
Also used in: none

Tangent & Normal

Question 47

How to find equation of tangent for a parabola if point of tangency is given?

Model Answer

$$T = 0: \quad yy_1 = 2a(x + x_1)$$

Any 2nd degree curve: replace $x^2 \to xx_1$, $y^2 \to yy_1$, $2x \to x + x_1$, $2y \to y + y_1$, $2xy \to xy_1 + x_1y$. Ex: $x^2 = y - 6$ at $(1, 7)$: $x = \frac{y+7}{2} - 6$, i.e. $y = 2x + 5$.

Grade - High

Practice map, main concept: Sheet: J-I 25-26; J-II 25, 65-66, 70-72; J-M 12, 16, 23, 52; J-A 24-25; C-I 14, 17; C-II 31 | DPP: 1(4); 6(13) | Question Bank: 81

Also used in: Sheet: J-I 33; J-M 1

Question 48

How to find equation of tangent for a parabola if any other condition other than point of tangency is given?

Model Answer

Take $y = mx + \frac{a}{m}$ (or $ty = x + at^2$) and use the given condition to find m (or t).

Ex: $y^2 = 16x$, parallel to $2x - y + 5 = 0$: $m = 2$, $y = 2x + 2$, contact $(1, 4)$.

Grade - Low

Practice map, main concept: Sheet: J-II 19

Also used in: DPP: 3(16)

Question 49

What is the equation of tangent at a point, whose coordinates are given in parametric form, on $y^2 = 4ax$?

Model Answer

$$ty = x + at^2$$

It meets the axis at $(-at^2, 0)$, the tangent at vertex at $(0, at)$ and the directrix at $\left(-a,\ a\left(t - \frac{1}{t}\right)\right)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 17, 20; J-II 21, 27; J-M 49; J-A 6, 10 | DPP: 6(11-12)

Also used in: Sheet: J-I 22; J-II 78; J-A 31; C-II 6-7 | DPP: 3(15); 4(12)

Question 50

How to find equation of tangent to a parabola $y^2 = 4ax$ passing through an external point?

Model Answer

Put (x_1, y_1) in $y = mx + \frac{a}{m}$: $x_1m^2 - y_1m + a = 0$; its two roots give the two tangents (or use $SS_1 = T^2$).

Ex: $y^2 = 12x$ from $(2, 5)$: $2m^2 - 5m + 3 = 0$, $m = 1, \frac{3}{2}$: $y = x + 3$, $3x - 2y + 4 = 0$.

Grade - High

Practice map, main concept: Sheet: J-II 20; J-M 2, 51, 53; J-A 37, 43; C-I 9 | DPP: 2(10); 4(7) | Question Bank: 87

Also used in: Sheet: J-II 65-66; J-M 39, 49; J-A 14, 33; C-I 17

Question 51
What is the relation between slope of tangent and parameter of point of tangency?

Model Answer

$$m = \frac{1}{t}$$

Grade - **Medium**

Practice map, main concept: Sheet: J-M 60; C-I 2
Also used in: Sheet: C-II 5

Question 52
How to find point of tangency if equation of tangent is given?

Model Answer

$$\left(\frac{a}{m^2}, \frac{2a}{m}\right)$$

Or compare the line with $yy_1 = 2a(x + x_1)$. Ex: $2x - y + 2 = 0$ on $y^2 = 16x$: $m = 2$, point $(1, 4)$.

Grade - **Medium**

Practice map, main concept: DPP: 3(4) | Question Bank: 86
Also used in: Sheet: J-II 19

Question 53
How to find equation of normal for a parabola if foot of normal is given?

Model Answer

$$y - y_1 = -\frac{y_1}{2a}(x - x_1)$$

(Line through the point perpendicular to the tangent.) Ex: $y^2 = 16x$ at $(16, 16)$: $y-16 = -2(x-16)$.

Grade - **Medium**

Practice map, main concept: Sheet: J-A 12, 20 | DPP: 3(10, 15); 4(11-12)
Also used in: Sheet: J-I 26; J-II 24; J-M 12 | DPP: 3(9, 11)

Question 54
Write expression for equation of normal to parabola $y^2 = 4ax$ in terms of parameter t.

Model Answer

$$y = -tx + 2at + at^3$$

It meets the axis at $G(2a + at^2, 0)$.

Grade - **Medium**

Practice map, main concept: Sheet: J-II 18, 22, 78, 89; J-A 40; C-II 12
Also used in: Sheet: J-I 18, 22-23; J-II 17; J-A 10

Question 55
Write expression for equation of normal to parabola $y^2 = 4ax$ in terms of slope m.

Model Answer

$$y = mx - 2am - am^3$$

Ex: $y^2 = 16x$, slope $-\frac{1}{4}$: $y = -\frac{x}{4} + 2 + \frac{1}{16}$, i.e. $4x + 16y = 33$.

Grade - Low

Practice map, main concept: DPP: 2(11); 3(5, 16)

Also used in: Sheet: J-II 39, 41

Question 56
What is the condition for a line to be a normal to parabola $y^2 = 4ax$?

Model Answer

$y = mx + c$ is a normal $\iff c = -2am - am^3$

Ex: $x + y = k$ normal to $y^2 = 12x$: $m = -1$, $k = c = 6 + 3 = 9$.

Grade - Low

Practice map, main concept: DPP: 3(2, 6)

Also used in: none

Question 57
What is the relation between slope of normal and parameter of foot of normal?

Model Answer

$$m = -t$$

Grade - Medium

Practice map, main concept: DPP: 3(8)

Also used in: Sheet: J-A 40; C-I 3 | DPP: 2(13)

Question 58
How to find foot of normal if equation of normal is given?

Model Answer

$$(am^2,\ -2am)$$

m = slope of the given normal (since $t = -m$).

Grade - Low

Practice map, main concept: none

Also used in: none

Question 59
What is point of intersection of tangents drawn at $P(t_1)$ and $Q(t_2)$ on parabola?

Model Answer

$$\big(at_1t_2,\ a(t_1 + t_2)\big)$$

(G.M. and A.M. of the abscissae/ordinates.) Ex: $y^2 = 4x$, $t = 1, 3$: $(3, 4)$.

Grade - Low

Practice map, main concept: Sheet: J-II 26; C-I 16; C-II 5, 7, 20

Also used in: none

Question 60
What is point of intersection of normals drawn at $P(t_1)$ and $Q(t_2)$ on parabola?

Model Answer

$$\left(2a + a(t_1^2 + t_1t_2 + t_2^2),\ -at_1t_2(t_1 + t_2)\right)$$

Grade - Low

Practice map, main concept: Sheet: J-II 17
Also used in: Sheet: J-II 26; C-II 11

Question 61
If normal to the parabola $y^2 = 4ax$ at $P(t_1)$ intersects parabola again at $Q(t_2)$, then what is the relation between t_1 and t_2?

Model Answer

$$t_2 = -t_1 - \frac{2}{t_1}$$

Ex: $y^2 = 8x$ at $(2, 4)$ $(t_1 = 1)$: $t_2 = -3$, so $Q(18, -12)$.

Grade - Medium

Practice map, main concept: Sheet: J-I 18-19, 21; J-II 23-24; J-M 1, 15; C-I 3; C-II 6, 16 | DPP: 2(13); 3(7, 11)
Also used in: Sheet: C-II 20 | DPP: 2(11)

Question 62
If normals to the parabola $y^2 = 4ax$ at $P(t_1)$ and $Q(t_2)$ intersect parabola again at $R(t_3)$, then what is the relation between t_1, t_2 and t_3?

Model Answer

$$t_1t_2 = 2, \qquad t_3 = -(t_1 + t_2)$$

Grade - Low

Practice map, main concept: Sheet: C-II 11, 18
Also used in: Sheet: C-I 16

Question 63
What is the angle between tangents drawn at ends of focal chord?

Model Answer

$$90^\circ \quad (m_1m_2 = \tfrac{1}{t_1t_2} = -1)$$

Grade - Medium

Practice map, main concept: Sheet: J-M 30 | DPP: 4(13)
Also used in: none

Question 64
Where do tangents drawn at ends of focal chord meet?

Model Answer
On the directrix, at $\left(-a,\ a\left(t - \frac{1}{t}\right)\right)$; for the latus rectum at the foot of the directrix $(-a, 0)$.

Grade - Medium

Practice map, main concept: none
Also used in: Sheet: J-M 12, 27; J-A 4, 33; C-II 17

Question 65

How to find minimum distance between a point and standard parabola

Model Answer

It is along the normal: make $y = -tx + 2at + at^3$ pass through the point, solve for t, and take the nearest foot $(at^2, 2at)$.

Ex: $(2, 8)$ and $y^2 = 4x$: $8 = t^3$, $t = 2$, foot $(4, 4)$, $d^2 = 20$.

Grade - Medium

Practice map, main concept: Sheet: J-I 24; J-A 18, 34-35 | DPP: 4(2) | Question Bank: 77, 80

Also used in: none

Question 66

How to find minimum distance between a line and standard parabola

Model Answer

If the line does not meet the parabola, take the tangent parallel to it: point $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$; distance from this point to the line (zero if they meet).

Ex: $y^2 = 4x$ and $y = x + 3$: point $(1, 2)$, distance $\frac{|1-2+3|}{\sqrt{2}} = \sqrt{2}$.

Grade - Low

Practice map, main concept: Sheet: J-I 32

Also used in: none

Question 67

What is the position of a line wrt parabola corresponding to number of common points between them?

Model Answer

$$m^2x^2 + 2(mc - 2a)x + c^2 = 0$$

$D > 0$: secant (2 points); $D = 0$ ($c = \frac{a}{m}$): tangent; $D < 0$: no common point. A line parallel to the axis meets the parabola once but is not a tangent.

Grade - Low

Practice map, main concept: Sheet: J-II 88 | DPP: 2(4)

Also used in: Sheet: C-II 15

Normal to Parabola from a point

Question 68

How many normals can be drawn from a point to a parabola?

Model Answer

$$am^3 + (2a - h)m + k = 0$$

A cubic in m: at most three normals from (h, k) (at least one real). Three real normals from $(h, 0)$ need $h > 2a$.

Ex: from $(15, 12)$ to $y^2 = 4x$: $m^3 - 13m + 12 = 0$, $m = 1, 3, -4$.

Grade - Medium

Practice map, main concept: Sheet: J-II 39-40; J-M 27 | DPP: 3(9)

Also used in: Sheet: J-II 99; C-II 14, 21, 56

Question 69
Define co-normal points for a parabola.

Model Answer
The feet of the normals drawn from one point. With slopes m_i (feet $t_i = -m_i$): $\sum m_i = 0$, $\sum m_i m_j = \frac{2a-h}{a}$, $m_1 m_2 m_3 = -\frac{k}{a}$; so $t_1 + t_2 + t_3 = 0$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 23; J-II 99; C-II 14 | DPP: 3(1, 12)
Also used in: none

Question 70
Where does the centroid of triangle, formed by three co-normal points on parabola, lie?

Model Answer
On the axis: ordinates $-2am_i$ add to zero. Centroid $\left(\frac{2(h-2a)}{3},\ 0\right)$.

Ex: normals from $(15, 12)$ to $y^2 = 4x$: centroid $\left(\frac{26}{3}, 0\right)$.

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 4(14)

Question 71
How to find locus of a point P if a condition on the slopes of the normals drawn from P to the parabola is given?

Model Answer
Use the cubic $am^3 + (2a - h)m + k = 0$ with its root relations; find the remaining root from the condition, put it in the cubic and replace (h, k) by (x, y).

Ex: two normals perpendicular: $m_1 m_2 = -1 \Rightarrow m_3 = \frac{k}{a}$, locus $y^2 = a(x - 3a)$.

Grade - **Low**
Practice map, main concept: Sheet: C-I 19; C-II 21, 56, 58
Also used in: Sheet: J-II 40

Question 72
How to find equation of circumcircle of triangle formed by three co-normal points wrt a point (h, k) for parabola $y^2 = 4ax$?

Model Answer

$$2(x^2 + y^2) - 2(h + 2a)x - ky = 0$$

It always passes through the vertex. Ex: $(15, 12)$, $y^2 = 4x$: $x^2 + y^2 - 17x - 6y = 0$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 51 | DPP: 4(14)
Also used in: Sheet: J-II 99

Rules for Transformation

Question 73
What are the rules for transformation to find results for other three standard parabolas and shifted parabolas?

Model Answer
$y^2 = -4ax$: replace a by $-a$. $x^2 = 4ay$: interchange x and y. $x^2 = -4ay$: interchange, then $a \to -a$. Shifted: replace x by $x - h$ and y by $y - k$.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-I 32

Question 74
How to find condition of tangency for other three standard parabolas and shifted parabolas?

Model Answer
For $y = mx + c$: $y^2 = -4ax$: $c = -\frac{a}{m}$; $x^2 = 4ay$: $c = -am^2$; $x^2 = -4ay$: $c = am^2$. Shifted $(y-k)^2 = 4a(x-h)$: $y - k = m(x-h) + \frac{a}{m}$.
Ex: $x + y = k$ touches $y = x - x^2$ when $k = 1$.

Grade - High
Practice map, main concept: Sheet: J-M 26 | DPP: 2(8); 3(3)
Also used in: Sheet: J-I 25; J-II 25, 92; J-M 10, 48, 53; J-A 28-29; C-II 8 | DPP: 4(1) | Question Bank: 86-87

Question 75
How to find condition for a line to be normal for other three standard parabolas and shifted parabolas?

Model Answer
For $y = mx + c$: $y^2 = -4ax$: $c = 2am + am^3$; $x^2 = 4ay$: $c = 2a + \frac{a}{m^2}$; $x^2 = -4ay$: $c = -2a - \frac{a}{m^2}$. Shifted: $y - k = m(x-h) - 2am - am^3$.

Grade - Medium
Practice map, main concept: Sheet: J-A 44 | DPP: 3(13)
Also used in: Sheet: J-II 40, 42; C-I 19 | DPP: 3(1)

Common Tangents

Question 76
How to find condition if a line is tangent/normal to one second degree curve and tangent/normal to another second degree curve?

Model Answer
Write the line in slope form for the first curve (e.g. $y = mx + \frac{a}{m}$), apply the tangency/normal condition of the second curve and solve for m.
Ex: $x^2 + y^2 = 2a^2$ and $y^2 = 8ax$: $y = mx + \frac{2a}{m}$, $\frac{4a^2}{m^2(1+m^2)} = 2a^2 \Rightarrow m = \pm 1$, $y = \pm(x + 2a)$.

Grade - High
Practice map, main concept: Sheet: J-I 35; J-II 41-42, 92, 94; J-M 10, 24, 32, 48; J-A 8, 26, 28-29, 41; C-I 7; C-II 19 | DPP: 4(1, 3); 6(3); 7(8-9); 10(7)
Also used in: Sheet: J-II 97; J-M 21

Question 77
What to do to minimize calculations in questions where a line is tangent/normal to one second degree curve and tangent/normal to another second degree curve?

Model Answer
Start from the curve whose slope form is simplest (parabola: $y = mx + \frac{a}{m}$), use the other condition only once, avoid points of contact until the end, and use symmetry (e.g. curves symmetric about $y = x$).

Grade - Low
Practice map, main concept: none
Also used in: none

Common Concepts

Question 78
Write expression for equation of chord whose mid point is given for $y^2 = 4ax$.

Model Answer

$$T = S_1 : \quad yy_1 - 2a(x + x_1) = y_1^2 - 4ax_1$$

Ex: chord of $y^2 = x$ bisected at $(2, 1)$: $x = 2y$, length $2\sqrt{5}$.

Grade - Medium
Practice map, main concept: Sheet: J-A 21 | DPP: 4(6, 8) | Question Bank: 90
Also used in: Sheet: J-II 49; C-II 18

Question 79
Write expression for equation of chord of contact for $y^2 = 4ax$.

Model Answer

$$T = 0 : \quad yy_1 = 2a(x + x_1)$$

For an external point (x_1, y_1).

Grade - Medium
Practice map, main concept: Sheet: J-II 45, 97; C-II 10 | DPP: 4(4)
Also used in: Sheet: J-A 43 | DPP: 2(10)

Question 80
How to find mid point of a chord of a standard parabola whose equation is given?

Model Answer
Compare the chord with $T = S_1$; quickly: slope $m = \frac{2a}{k}$ gives $k = \frac{2a}{m}$, then h from the chord.
Ex: $2x + y = 2$ on $y^2 = 16x$: $k = \frac{8}{-2} = -4$, $h = 3$.

Grade - Medium
Practice map, main concept: none
Also used in: Sheet: J-A 21

Question 81
How to find external point for a chord of contact if equation of chord of contact is given?

Model Answer
Compare $lx + my + n = 0$ with $yy_1 - 2ax - 2ax_1 = 0$: point $\left(\frac{n}{l},\ -\frac{2am}{l}\right)$ (the pole of the line).

Grade - **Low**
Practice map, main concept: none
Also used in: DPP: 4(4)

Question 82
What is formula for length of chord of contact, wrt (x_1, y_1), of standard parabola?

Model Answer

$$L = \frac{\sqrt{(y_1^2 - 4ax_1)(y_1^2 + 4a^2)}}{a}$$

Ex: $(-1, 2)$, $y^2 = 4x$: $L = \sqrt{8 \cdot 8} = 8$.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 83
What is formula for area formed by tangents drawn from (x_1, y_1) to standard parabola and corresponding chord of contact?

Model Answer
Area $= \frac{(y_1^2 - 4ax_1)^{3/2}}{2a}$
Ex: $(-1, 2)$, $y^2 = 4x$: $\frac{8^{3/2}}{2} = 8\sqrt{2}$.

Grade - **Low**
Practice map, main concept: Sheet: J-II 64
Also used in: Sheet: J-II 45

Question 84
Write expression for combined equation of pair of tangents for $y^2 = 4ax$.

Model Answer

$$SS_1 = T^2 :\ (y^2 - 4ax)(y_1^2 - 4ax_1) = \{yy_1 - 2a(x + x_1)\}^2$$

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 85
Write expression for equation of polar for $y^2 = 4ax$.

Model Answer

$$T = 0 :\quad yy_1 = 2a(x + x_1)$$

Polar of the focus is the directrix.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 86
What does $T = 0$ represent for a point, according to its position relative to parabola?

Model Answer
On the parabola: the tangent; outside: the chord of contact; inside: a line not meeting the parabola. In every case it is the polar.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 87
How to find coordinates of pole of a given line?

Model Answer

$$lx + my + n = 0 \;\Rightarrow\; \left(\frac{n}{l},\; -\frac{2am}{l}\right)$$

(Compare with $T = 0$.)

Grade - Low
Practice map, main concept: none
Also used in: none

Question 88
From any variable point on a fixed line, tangents are drawn to the parabola then chord of contact will pass through which point?

Model Answer
Through the pole of the fixed line, $\left(\frac{n}{l},\; -\frac{2am}{l}\right)$.
Ex: line $x - y + 3 = 0$, $y^2 = 8x$: fixed point $(3, 4)$.

Grade - Low
Practice map, main concept: DPP: 4(5)
Also used in: none

Diameter

Question 89
Define diameter for a second degree curve.

Model Answer
Locus of the mid-points of a system of parallel chords.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 90
Derive equation of diameter for standard parabola.

Model Answer

$$y = \frac{2a}{m}$$

From $T = S_1$ the slope of the chord with mid-point (h, k) is $\frac{2a}{k} = m$. Every diameter is parallel to the axis. Ex: chords of slope 2 of $y^2 = 8x$: $y = 2$.

Grade - Low

Practice map, main concept: none
Also used in: none

Highlights

Question 91

What angle does portion of tangent intercepted between directrix and point of tangency subtend at focus?

Model Answer

$$90^\circ$$

Grade - Medium

Practice map, main concept: Sheet: J-A 33
Also used in: none

Question 92

Where does foot of perpendicular from focus on any tangent to parabola lies?

Model Answer

On the tangent at the vertex ($x = 0$ for $y^2 = 4ax$).

Grade - Low

Practice map, main concept: Sheet: J-II 52; C-II 17
Also used in: Sheet: J-I 17

Question 93

What is harmonic mean of segments of any focal chord of parabola?

Model Answer

$$\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$$

So the H.M. of SP and SQ is $2a$ = semi latus rectum.

Ex: segments 2 and 4 for $y^2 = 8ax$: $\frac{1}{2} + \frac{1}{4} = \frac{1}{2a}$, $a = \frac{2}{3}$.

Grade - Medium

Practice map, main concept: DPP: 4(10) | Question Bank: 83
Also used in: Question Bank: 71

Question 94

What is the relation between directrix and circle drawn on any focal chord as diameter?

Model Answer

The circle touches the directrix.

Grade - Low

Practice map, main concept: none
Also used in: Sheet: C-I 15

Question 95
What is the relation between tangent at vertex and circle drawn on any focal distance as diameter?

Model Answer
The circle touches the tangent at the vertex.

Grade - Medium
Practice map, main concept: Question Bank: 70
Also used in: none

Question 96
If tangent and normal drawn at P on parabola intersect the axis at T and G respectively, then what is the circumcentre of triangle TPG?

Model Answer
The focus S: $ST = SG = SP$ and S is the mid-point of TG.

Grade - Low
Practice map, main concept: Sheet: J-I 22
Also used in: Sheet: C-II 62

Question 97
Explain reflection property of parabola.

Model Answer
The tangent at P bisects the angle between SP and the perpendicular PM to the directrix; so rays from S reflect parallel to the axis, and rays parallel to the axis reflect through S.

Grade - Low
Practice map, main concept: none
Also used in: none

Question 98
What is the ratio of area of triangle formed by any 3 points on parabola to area of triangle formed by tangents at these 3 points?

Model Answer

$$2 : 1$$

Area of triangle $t_1, t_2, t_3 = a^2|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)|$; triangle of tangents: half of it.

Grade - Low
Practice map, main concept: Sheet: J-II 53
Also used in: none

Question 99
Where does image of focus wrt any tangent to the parabola lies?

Model Answer
On the directrix.

Grade - Low
Practice map, main concept: none
Also used in: Sheet: J-II 52